

Project Overview:

Evaluation of the MyCourses Feedback Tool in Small Courses: Inspection-Based and User-Based Teacher's Perspectives

The project evaluated a course feedback tool (an LTI plugin /Spark/Studio SaaS) integrated into Aalto University's MyCourses learning environment, focusing specifically on the perspective of instructors teaching small courses (fewer than 50 students). As the tool is used only a few times a year, the system's learnability and memorability are subject to exceptionally high demands.

The evaluation was conducted as a two-phase study combining expert evaluation and user testing. My responsibilities included developing the research plan, recruiting users, preparing testing equipment and materials (including screen and audio recordings and SUS questionnaires), facilitating the tests, and analyzing and reporting both quantitative and qualitative data.

What did I do during the project?

Method selection and research design:

- We chose the Cognitive Walkthrough method over heuristic evaluation because it specifically addresses learnability and cognitive barriers for occasional users.
- We defined six standardized test scenarios based on the instructor's core workflows: setting course dates, launching the tool, creating a multiple-choice question, utilizing the question bank, activating the consent prompt, and downloading the list of respondents.
- Expert evaluation (inspection-based): We each independently conducted a cognitive walkthrough based on four key questions and rated the severity of our findings using Nielsen's severity scale (0–4).

User-based testing:

- I designed and piloted the testing setup and subsequently conducted user tests with three actual Aalto University course coordinators and instructors.

The tests utilized:

- The think-aloud method (verbalizing thoughts while performing tasks)
- Recording of screen activity, mouse movements, and audio
- The standard System Usability Scale (SUS) questionnaire

- A semi-structured post-test interview to gain deeper qualitative feedback.
- Analysis and triangulation of results: I combined observational data, task completion times, error rates, and SUS scores into a unified synthesis.

Stakeholder communication and concrete development proposals:

- We translated our findings into concrete UI/UX improvement proposals (e.g., auto-save, contextual tooltips, a "What's New" changelog) and communicated them to the relevant stakeholders to support further development.

Key findings and insights

- Functionality does not guarantee usability (SUS: 46.7 / 100):
The tool performs the required technical functions, yet its cognitive load is unsustainable. The average SUS score of 46.7 corresponds to a grade of "F" (placing it in the bottom 15% of standardized systems). Teachers were constantly forced to rely on trial and error, leading to frustration and uncertainty regarding whether their data had been successfully saved.
- Seasonal use turns even experienced users into novices: My most significant insight was how poor learnability and memorability negate user routine. Because the feedback tool is used only at the end of a course, even experienced teachers operate like first-time users. This demonstrated that in enterprise and administrative systems, clear visual guidance, standardized terminology, and real-time system feedback (e.g., a clear "Saved" status rather than a button hidden at the bottom of the page) are critical.
- Project management and professional resilience: Navigating project team challenges, recruiting users under a tight schedule, and bearing responsibility for the functionality of the entire testing setup and data fostered independent problem-solving skills and the ability to work under pressure. I learned to take charge of the situation, communicate professionally with stakeholders, and see a high-quality research project through to completion.